



DC – AI – Routing – Switching – Security – Wireless – Automation

## **L3VPN over SRv6 with JCNr**

An HPE Juniper Networking Techpost

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2026-01-11

**HPE**



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# 1 L3VPN over SRv6 with JCNR

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Do you need secure, isolated multi-tenant connectivity across Kubernetes and cloud infrastructures. JCNR supports SRv6 L3VPN with micro-Segment Identifiers (uSIDs) in various SRv6 endpoint behaviors (End.DT4, End.DT6, End.DT46).

## 1.1 Overview

Juniper Cloud-Native Router (JCNR) is a containerized, cloud-native routing solution that brings enterprise-grade networking capabilities to cloud and containerized environments. Built on Juniper's proven routing technologies, JCNR delivers the same robust features and characteristics as traditional Juniper routers while being optimized for modern cloud-native infrastructures.

Segment Routing over IPv6 (SRv6) is a modern networking paradigm that leverages the IPv6 protocol to provide advanced traffic engineering, service programming, and network simplification. SRv6 combines the benefits of segment routing with the ubiquity of IPv6 enabling sophisticated network services through native IPv6 packet processing.

In this article, we will demonstrate SRv6 and SRv6 L3VPN solution offered by JCNR. The feature is supported starting from Junos 24.2.

## 1.2 SRv6 and L3VPN

Layer 3 Virtual Private Networks (L3VPNs) represent a fundamental service model in modern networking that enables organizations to establish secure, isolated communication channels across shared infrastructure. This enables admins to host multiple customers on a shared infrastructure for better resource utilization and yet providing services.

SRv6 L3VPN leverages IPv6 segment routing to provide Layer 3 VPN services across cloud-native network infrastructures. Unlike traditional MPLS-based L3VPN implementations, SRv6 utilizes IPv6 addresses as segment identifiers, enabling service programming directly within the IPv6 header with and without Segment Routing Header (SRH) extension.

## 1.3 L3VPN with JCNR

JCNR supports l3vpn service with SR-MPLS as ingress, transit and egress nodes in the MPLS network. L3VPN service is supported with SRv6 as ingress and egress node. JCNR offers carrier grade routing functionality with support for IS-IS and OSPF as IGP while BGP offering VPN functionality in control plane.

JCNR offers multiple customers hosted on the same instance with flexibility in attaching users on the fly like other JUNOS platforms. Along with this, JCNR support L3VPN solution in CNI mode where a PoDs running on the same Kubernetes cluster can be hosted on a VPN instance providing connectivity to the service to its end points. All of this is also offered with IPSec service when service chained with cSRX, if user desired security along with L3VPN with SRv6.

## 1.4 SRv6 with JCNR

Segment Routing over IPv6 (SRv6) in Juniper Cloud-Native Router (JCNR) is supported with micro segment identifiers (uSID). Micro-Segment Identifiers (uSIDs) represent a significant optimization technique in SRv6 that addresses the header overhead concerns of traditional SRv6 implementations. While standard SRv6 uses full 128-bit IPv6 addresses as segment identifiers, uSIDs enable the packing of multiple segments within a single IPv6 address, dramatically reducing the Segment Routing Header (SRH) size and improving network efficiency. In some cases, removing the need for SRH header as uSIDs fit in destination address of IPv6 header. JCNR supports max of 6 uSIDs which requires no SRH header addition.

SRv6 combines the benefits of segment routing with the native IPv6 forwarding plane providing enhanced service programming capabilities through segment identifier (SID) functions. JCNR supports SRv6 head-end and egress nodes roles in a SRv6 network. Support for transit node is in the pipeline.

SRv6 has multiple end point behaviors. In JCNR, following end points are supported.

- **End.DT4:** Endpoint with decapsulation and specific IPv4 table lookup function for SRv6 instantiation of Global or IPv4 L3VPN (transport IPv4 services over SRv6 underlay)
- **End.DT6:** Endpoint with decapsulation and specific IPv6 table lookup function for SRv6 instantiation of Global or IPv6 L3VPN (Transport IPv6 services over SRv6 underlay)
- **End.DT46:** Endpoint with decapsulation and specific IP table lookup function for SRv6 instantiation of Global, IPv4 or IPv6 L3VPN (Transport both IPv4 and IPv6 services over SRv6 underlay). It is shared across IPv4 and IPv6 prefixes.

JCNR functions as a comprehensive Container Network Interface (CNI) plugin for Kubernetes, providing advanced networking capabilities directly integrated with the container orchestration platform. JCNR operates also as a Cloud-Native Network Function (CNF), providing traditional network services (routing, switching, security) in containerized form. As a CNF, JCNR delivers carrier-grade network functions with cloud-native operational characteristics. SRv6 is supported in both CNI and CNF modes giving the flexibility for users to deploy SRv6 solution in any cloud native environments. JCNR supports up to 6 uSIDs.

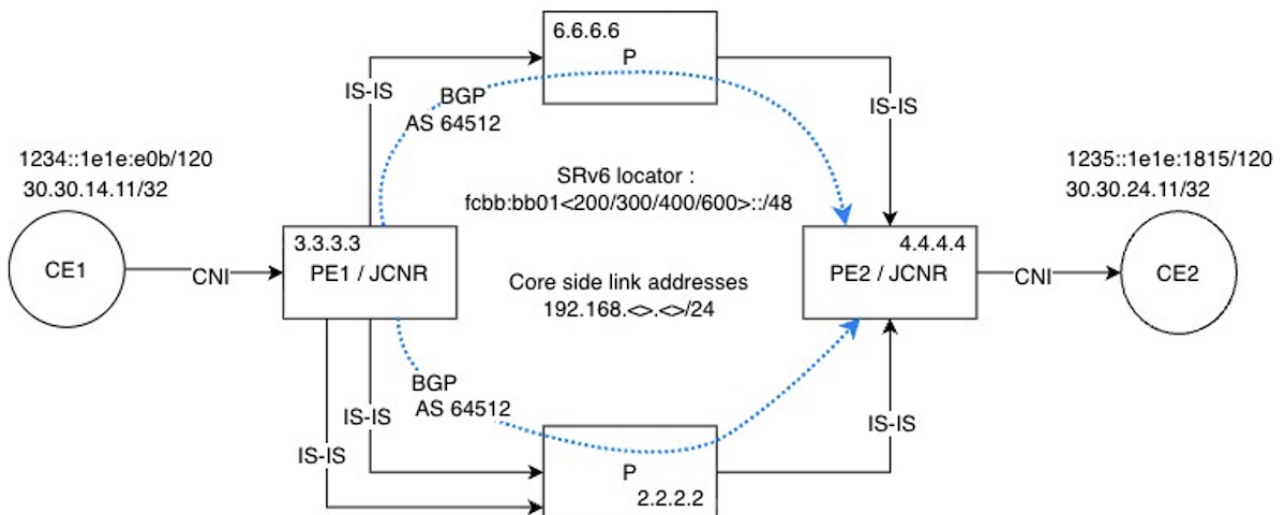
## 1.5 Solution: L3VPN over SRv6 with JCNR

In this document, we will demonstrate the following solution with JCNR. In this topology, PE1 and PE2 are JCNR acting as SRv6 head-end and egress nodes in the network. There is a BGP session between PE1 and PE2 for providing VPN solution with AS 64512. IS-IS is used as IGP. There is a redundant path between PE1 and provide node to demonstrate ECMP support with SRv6.

CE1 and CE2 in this solution are PoDs connected to JCNR. JCNR is acting as secondary CNI in this solution demonstrating the ability to provide SRv6 connectivity in CNI mode.

In the context of 5G and Open RAN (O-RAN) network architecture, the midhaul is the transport network link that connects the Distributed Unit (DU) to the Centralized Unit (CU). JCNR with SRv6 can be used as gateway for DU where JCNR is providing connectivity over SRv6 network in midhaul to CU. In the same use case, JCNR can be the gateway for CU providing connectivity to multiple DUs.

### 1.5.1 Topology for this Solution



**Figure 1:** Lab Topology

### 1.5.2 Kubernetes Cluster from PE1

```

1 jcnr3@jcnr3-kvm:~/srv6$ kubectl get pods -A
2 NAMESPACE NAME READY STATUS RESTARTS AGE
3 calico-apiserver calico-apiserver-58b6c8b8ff-5wprj 1/1 Running 1 83d
4 calico-apiserver calico-apiserver-58b6c8b8ff-wjck 1/1 Running 1 83d
5 calico-system calico-kube-controllers-78788579b8-88nd9 1/1 Running 1 83d
6 calico-system calico-node-gsf84 1/1 Running 1 83d
7 calico-system calico-typha-6b56d9c9db-6rxmm 1/1 Running 2 (21d ago) 83d
8 contrail-deploy contrail-k8s-deployer-7b8c565984-p2jmd 1/1 Running 0 21d
9 contrail-deploy jcnr-config-controller-m8w9x 1/1 Running 2 (21d ago) 21d
10 contrail contrail-tools-9gtlh 1/1 Running 0 21d
11 contrail jcnr-0-contrail-vrouter-nodes-szklq 2/2 Running 7 (11d ago) 21d
12 contrail jcnr-0-contrail-vrouter-nodes-vrdpdk-mkrqx 1/1 Running 5 (11d ago) 21d
13 jcnr jcnr-0-crpd-0 2/2 Running 0 11d
14 jcnr syslog-ng-jcnrobj-kr5rm 1/1 Running 0 21d
15 kube-system coredns-5dd5756b68-gbmsx 1/1 Running 1 83d
16 kube-system coredns-5dd5756b68-xzwj9 1/1 Running 1 83d
17 kube-system etcd-jcnr3-kvm 1/1 Running 29 83d
18 kube-system kube-apiserver-jcnr3-kvm 1/1 Running 1 83d
19 kube-system kube-controller-manager-jcnr3-kvm 1/1 Running 1 83d
20 kube-system kube-multus-ds-prcbm 1/1 Running 1 83d
21 kube-system kube-proxy-qn4wx 1/1 Running 1 83d
22 kube-system kube-scheduler-jcnr3-kvm 1/1 Running 1 83d
23 tigera-operator tigera-operator-6fbc4f6f8d-228zt 1/1 Running 2 (21d ago) 83d
24 srv6 CE1 1/1 Running 0 153m

```

Protocol and data path state from PE1 and PE2 from this topology given below. In this topology CE1 and CE2 are connected to JCNr as PoDs in CNI mode. Yet, they are going over SRv6 network as VPN traffic. This demonstrates the advantage of JCNr as SRv6 node in cloud native environments.

### 1.5.3 IS-IS state on PE1

```

1 root@jcnr3-kvm# run show isis adjacency
2 Interface System L State Hold (secs) SNPA
3 enp10s0 jcnr2 2 Up 587
4 enp7s0 jcnr2 2 Up 587
5 enp9s0 jcnr6-kvm 2 Up 591

```

### 1.5.4 IS-IS state on PE2

```

1 root@jcnr4-kvm# run show isis adjacency
2 Interface System L State Hold (secs) SNPA
3 enp10s0 jcnr6-kvm 2 Up 599
4 enp8s0 jcnr2 2 Up 591

```

### 1.5.5 BGP state on PE1

```

1 root@jcnr3-kvm# run show bgp summary
2 Threading mode: BGP I/O
3 TCP listen port: 178
4 Default eBGP mode: advertise - accept, receive - accept
5 Groups: 2 Peers: 2 Down peers: 0
6 Table Tot Paths Act Paths Suppressed History Damp State Pending
7 inet.0
8         0 0 0 0 0 0
9 inet6.0
10        0 0 0 0 0 0
11 bgp.l3vpn.0
12        1 1 0 0 0 0
13 bgp.l3vpn-inet6.0
14        1 1 0 0 0 0
15 bgp.evpn.0
16        0 0 0 0 0 0
17 Peer AS InPkt OutPkt OutQ Flaps Last Up/Dwn State|#Active/Received/Accepted/Damped...
18 4.4.4.4 64512 241 244 0 0 1:40:11 Establ
19   inet.0: 0/0/0/0
20   inet6.0: 0/0/0/0
21   bgp.l3vpn.0: 1/1/1/0
22   bgp.l3vpn-inet6.0: 1/1/1/0
23   bgp.evpn.0: 0/0/0/0
24   srv6.inet.0: 1/1/1/0
25   srv6.inet6.0: 1/1/1/0

```

### 1.5.6 BGP state on PE2

```

1 root@jcnr4-kvm# run show bgp summary
2 Threading mode: BGP I/O
3 TCP listen port: 178
4 Default eBGP mode: advertise - accept, receive - accept
5 Groups: 2 Peers: 2 Down peers: 0
6 Table Tot Paths Act Paths Suppressed History Damp State Pending
7 inet.0
8         0 0 0 0 0 0
9 inet6.0
10        0 0 0 0 0 0
11 bgp.l3vpn.0
12        1 1 0 0 0 0
13 bgp.l3vpn-inet6.0
14        1 1 0 0 0 0
15 bgp.evpn.0
16        0 0 0 0 0 0
17 Peer AS InPkt OutPkt OutQ Flaps Last Up/Dwn State|#Active/Received/Accepted/Damped...
18 3.3.3.3 64512 244 238 0 0 1:39:32 Establ
19   inet.0: 0/0/0/0
20   inet6.0: 0/0/0/0
21   bgp.l3vpn.0: 1/1/1/0
22   bgp.l3vpn-inet6.0: 1/1/1/0
23   bgp.evpn.0: 0/0/0/0
24   srv6.inet.0: 1/1/1/0
25   srv6.inet6.0: 1/1/1/0

```

### 1.5.7 Local SID information on PE1

```

1 root@jcnr3-kvm> show srv6 local-sids
2 SID SID-Owner SID-Type Locator SID-Behavior
3 fcbb:bb01:300:e000:: BGP DYNAMIC u_loc End.DT4 with NEXT-CSID
4 fcbb:bb01:300:e001:: BGP DYNAMIC u_loc End.DT6 with NEXT-CSID
5 fcbb:bb01:300:e002:: BGP DYNAMIC u_loc End.DT46 with NEXT-CSID

```

```

6 fcbb:bb01:300:e003:: BGP DYNAMIC u_loc End.DT4 with NEXT-CSID
7 fcbb:bb01:300:e004:: BGP DYNAMIC u_loc End.DT6 with NEXT-CSID
8 fcbb:bb01:300:e005:: BGP DYNAMIC u_loc End.DT46 with NEXT-CSID
9
10 root@jcnr3-kvm> show srv6 locator
11 Locator: u_loc
12 Locator prefix: fcbb:bb01:300::, Locator length: 48
13 Block length: 32, Node length: 16
14 Function length: 16, Argument length: 0
15 Micro SID Locator, Flavor [ None ]
16 Micro SID Block Name: usid_blk_with_statics
17
18 root@jcnr3-kvm> show srv6 block
19 Block: usid_blk_with_statics
20 Block Prefix: fcbb:bb01::, Block length: 32, Micro-sid length: 16
21 Global Micro SIDs:
22 Static SID range: 0x0-0xDFFF, Dynamic SID range: -
23 Allocated static SID count: 1, Allocated dynamic SID count: 0
24 Available static SID count: 57343, Available dynamic SID count: 0
25 Local Micro SIDs:
26 Static SID range: 0xF830-0xFFFF, Dynamic SID range: 0xE000-0xF82F
27 Allocated static SID count: 0, Allocated dynamic SID count: 6
28 Available static SID count: 2000, Available dynamic SID count: 6186

```

### 1.5.8 BGP Peer route on PE1

```

1 root@jcnr3-kvm> show route 4.4.4.4 detail
2
3 inet.0: 19 destinations, 22 routes (19 active, 0 holddown, 0 hidden)
4 4.4.4.4/32 (1 entry, 1 announced)
5 *IS-IS Preference: 18
6 Level: 2
7 Next hop type: Router, Next hop index: 0
8 Address: 0x58f199acf61c
9 Next-hop reference count: 2, Next-hop session id: 0
10 Kernel Table Id: 0
11 Next hop: 192.168.200.2 via enp10s0, selected
12 Session Id: 0
13 Next hop: 192.168.133.2 via enp7s0
14 Session Id: 0
15 Next hop: 192.168.155.6 via enp9s0
16 Session Id: 0
17 State: <Active Int>
18 Age: 23:24 Metric: 20
19 Validation State: unverified
20 ORR Generation-ID: 0
21 Task: IS-IS
22 Announcement bits (5): 1-KRT MFS 2-KRT 3-Resolve tree 3 6-KRT-vRouter 8-Resolve tree
23 1
24 AS path: I
25 Thread: junos-main

```

### 1.5.9 Remote VPN route on PE1

```

1 root@jcnr3-kvm# run show route 30.30.24.11/32 detail
2
3 srv6.inet.0: 3 destinations, 3 routes (3 active, 0 holddown, 0 hidden)
4 30.30.24.11/32 (1 entry, 1 announced)
5 *BGP Preference: 170/-101
6 Route Distinguisher: 10.87.3.248:2
7 Next hop type: Indirect, Next hop index: 0
8 Address: 0x634bd311143c
9 Next-hop reference count: 4
10 Kernel Table Id: 0
11 Source: 4.4.4.4
12 Next hop type: List, Next hop index: 0
13 Next hop: ELNH Address 0x634bd311151c, selected
14 Next hop type: Chain, Next hop index: 0
15 Address: 0x634bd311151c
16 Next-hop reference count: 1, Next-hop session id: 0
17 Kernel Table Id: 0
18 Next hop: via Chain Tunnel Composite, SRv6 (src 3333::1 dest fcbb:bb01:400::)
19 Next hop: ELNH Address 0x634bd036e91c

```



```

20 SRV6-Tunnel: Reduced-SRH Encap-mode Remove-Last-Sid Propagate-CoS
21 Src: 3333::1 Dest: fcbb:bb01:400::
22 Segment-list[0] fcbb:bb01:400::
23 Gateway opaque handle: 0x634bd0201b60
24 Next hop type: Router, Next hop index: 0
25 Address: 0x634bd036e91c
26 Next-hop reference count: 9, Next-hop session id: 0
27 Kernel Table Id: 0
28 Next hop: fe80::5054:ff:fe4b:1643 via enp10s0
29 Next hop: ELNH Address 0x634bd037107c
30 Next hop type: Chain, Next hop index: 0
31 Address: 0x634bd037107c
32 Next-hop reference count: 1, Next-hop session id: 0
33 Kernel Table Id: 0
34 Next hop: via Chain Tunnel Composite, SRv6 (src 3333::1 dest fcbb:bb01:400::)
35 Next hop: ELNH Address 0x634bd03730dc
36 SRV6-Tunnel: Reduced-SRH Encap-mode Remove-Last-Sid Propagate-CoS
37 Src: 3333::1 Dest: fcbb:bb01:400::
38 Segment-list[0] fcbb:bb01:400::
39 Gateway opaque handle: 0x634bd0201b60
40 Next hop type: Router, Next hop index: 0
41 Address: 0x634bd03730dc
42 Next-hop reference count: 9, Next-hop session id: 0
43 Kernel Table Id: 0
44 Next hop: fe80::d8d3:faff:fe05:d38 via enp7s0
45 Next hop: ELNH Address 0x634bd0371cbc
46 Next hop type: Chain, Next hop index: 0
47 Address: 0x634bd0371cbc
48 Next-hop reference count: 1, Next-hop session id: 0
49 Kernel Table Id: 0
50 Next hop: via Chain Tunnel Composite, SRv6 (src 3333::1 dest fcbb:bb01:400::)
51 Next hop: ELNH Address 0x634bd037203c
52 SRV6-Tunnel: Reduced-SRH Encap-mode Remove-Last-Sid Propagate-CoS
53 Src: 3333::1 Dest: fcbb:bb01:400::
54 Segment-list[0] fcbb:bb01:400::
55 Gateway opaque handle: 0x634bd0201b60
56 Next hop type: Router, Next hop index: 0
57 Address: 0x634bd037203c
58 Next-hop reference count: 14, Next-hop session id: 0
59 Kernel Table Id: 0
60 Next hop: fe80::5054:ff:fe00:a914 via enp9s0
61 Protocol next hop: fcbb:bb01:400::
62 Composite next hop: 0x634bd09c2400 - INH Session ID: 0, CNH non-key opaque: (nil),
    CNH key opaque: 0x634bd09c2340
63 Indirect next hop: 0x634bd0564988 - INH Session ID: 0, INH non-key opaque: 0
    x634bd0207a40, INH key opaque: (nil)
64 State: <Secondary Active Int Ext OpaqueData ProtectionCand>
65 Peer AS: 64512
66 Age: 4:03 Metric2: 20
67 Validation State: unverified
68 ORR Generation-ID: 0
69 Task: BGP_64512_64512.4.4.4
70 Announcement bits (4): 2-KRT MFS 3-KRT 4-KRT-vRouter 6-PLFM-LAYER
71 AS path: I
72 Communities: target:64512:4
73 Import Accepted MultiNextHop RecvNextHopIgnored
74 SRv6 SID: fcbb:bb01:400:: Service tlv type: 5 Behavior: 63 BL: 32 NL: 16 FL: 16 AL:
    0 TL: 16 TO: 48
75 VPN Label: 917552
76 Localpref: 100
77 Router ID: 4.4.4.4
78 Primary Routing Table: bgp.l3vpn.0
79 Thread: junos-main
80
81 bgp.l3vpn.0: 1 destinations, 1 routes (1 active, 0 holddown, 0 hidden)
82
83 10.87.3.248:2:30.30.24.11/32 (1 entry, 0 announced)
84 *BGP Preference: 170/-101
85 Route Distinguisher: 10.87.3.248:2
86 Next hop type: Indirect, Next hop index: 0
87 Address: 0x634bd311143c
88 Next-hop reference count: 4
89 Kernel Table Id: 0
90 Source: 4.4.4.4
91 Next hop type: List, Next hop index: 0
92 Next hop: ELNH Address 0x634bd311151c, selected
93 Next hop type: Chain, Next hop index: 0
94 Address: 0x634bd311151c
95 Next-hop reference count: 1, Next-hop session id: 0
96 Kernel Table Id: 0
97 Next hop: via Chain Tunnel Composite, SRv6 (src 3333::1 dest fcbb:bb01:400::)

```

```

98     Next hop: ELNH Address 0x634bd036e91c
99     SRV6-Tunnel: Reduced-SRH Encap-mode Remove-Last-Sid Propagate-CoS
100     Src: 3333::1 Dest: fcbb:bb01:400::
101     Segment-list[0] fcbb:bb01:400::
102     Gateway opaque handle: 0x634bd0201b60
103     Next hop type: Router, Next hop index: 0
104     Address: 0x634bd036e91c
105     Next-hop reference count: 9, Next-hop session id: 0
106     Kernel Table Id: 0
107     Next hop: fe80::5054:ff:fe4b:1643 via enp10s0
108     Next hop: ELNH Address 0x634bd037107c
109     Next hop type: Chain, Next hop index: 0
110     Address: 0x634bd037107c
111     Next-hop reference count: 1, Next-hop session id: 0
112     Kernel Table Id: 0
113     Next hop: via Chain Tunnel Composite, SRv6 (src 3333::1 dest fcbb:bb01:400::)
114     Next hop: ELNH Address 0x634bd03730dc
115     SRV6-Tunnel: Reduced-SRH Encap-mode Remove-Last-Sid Propagate-CoS
116     Src: 3333::1 Dest: fcbb:bb01:400::
117     Segment-list[0] fcbb:bb01:400::
118     Gateway opaque handle: 0x634bd0201b60
119     Next hop type: Router, Next hop index: 0
120     Address: 0x634bd03730dc
121     Next-hop reference count: 9, Next-hop session id: 0
122     Kernel Table Id: 0
123     Next hop: fe80::d8d3:faff:fe05:d38 via enp7s0
124     Next hop: ELNH Address 0x634bd0371cbc
125     Next hop type: Chain, Next hop index: 0
126     Address: 0x634bd0371cbc
127     Next-hop reference count: 1, Next-hop session id: 0
128     Kernel Table Id: 0
129     Next hop: via Chain Tunnel Composite, SRv6 (src 3333::1 dest fcbb:bb01:400::)
130     Next hop: ELNH Address 0x634bd037203c
131     SRV6-Tunnel: Reduced-SRH Encap-mode Remove-Last-Sid Propagate-CoS
132     Src: 3333::1 Dest: fcbb:bb01:400::
133     Segment-list[0] fcbb:bb01:400::
134     Gateway opaque handle: 0x634bd0201b60
135     Next hop type: Router, Next hop index: 0
136     Address: 0x634bd037203c
137     Next-hop reference count: 14, Next-hop session id: 0
138     Kernel Table Id: 0
139     Next hop: fe80::5054:ff:fe00:a914 via enp9s0
140     Protocol next hop: fcbb:bb01:400::
141     Composite next hop: 0x634bd09c2400 - INH Session ID: 0, CNH non-key opaque: (nil),
        CNH key opaque: 0x634bd09c2340
142     Indirect next hop: 0x634bd0564988 - INH Session ID: 0, INH non-key opaque: 0
        x634bd0207a40, INH key opaque: (nil)
143     State: <Active Int Ext ProtectionPath ProtectionCand>
144     Peer AS: 64512
145     Age: 4:03 Metric2: 20
146     Validation State: unverified
147     ORR Generation-ID: 0
148     Task: BGP_64512_64512.4.4.4
149     AS path: I
150     Communities: target:64512:4
151     Import Accepted MultiNextHop RecvNextHopIgnored
152     SRv6 SID: fcbb:bb01:400:: Service tlv type: 5 Behavior: 63 BL: 32 NL: 16 FL: 16 AL:
        0 TL: 16 TO: 48
153     VPN Label: 917552
154     Localpref: 100
155     Router ID: 4.4.4.4
156     Secondary Tables: srv6.inet.0
157     Thread: junos-main

```

### 1.5.10 Remote VPN route on PE2 : Control plane

```

1  root@jcnr4-kvm# run show route 30.30.14.11/32 detail
2
3  srv6.inet.0: 3 destinations, 3 routes (3 active, 0 holddown, 0 hidden)
4  30.30.14.11/32 (1 entry, 1 announced)
5      *BGP Preference: 170/-101
6          Route Distinguisher: 10.87.3.233:2
7          Next hop type: Indirect, Next hop index: 0
8          Address: 0x5f6e06f7265c
9          Next-hop reference count: 4
10         Kernel Table Id: 0
11         Source: 3.3.3.3

```

```

12         Next hop type: List, Next hop index: 0
13     Next hop: ELNH Address 0x5f6e06f71a1c, selected
14         Next hop type: Chain, Next hop index: 0
15         Address: 0x5f6e06f71a1c
16         Next-hop reference count: 1, Next-hop session id: 0
17         Kernel Table Id: 0
18         Next hop: via Chain Tunnel Composite, SRv6 (src 4444::1 dest fcbb:bb01:300::)
19         Next hop: ELNH Address 0x5f6e06f715bc
20         SRV6-Tunnel: Reduced-SRH Encap-mode Remove-Last-Sid Propagate-CoS
21         Src: 4444::1 Dest: fcbb:bb01:300::
22         Segment-list[0] fcbb:bb01:300::
23         Gateway opaque handle: 0x5f6e06e04c20
24         Next hop type: Router, Next hop index: 0
25         Address: 0x5f6e06f715bc
26         Next-hop reference count: 14, Next-hop session id: 0
27         Kernel Table Id: 0
28         Next hop: fe80::5054:ff:feab:aecc via enp10s0
29     Next hop: ELNH Address 0x5f6e06f7655c
30         Next hop type: Chain, Next hop index: 0
31         Address: 0x5f6e06f7655c
32         Next-hop reference count: 1, Next-hop session id: 0
33         Kernel Table Id: 0
34         Next hop: via Chain Tunnel Composite, SRv6 (src 4444::1 dest fcbb:bb01:300::)
35         Next hop: ELNH Address 0x5f6e06f72b9c
36         SRV6-Tunnel: Reduced-SRH Encap-mode Remove-Last-Sid Propagate-CoS
37         Src: 4444::1 Dest: fcbb:bb01:300::
38         Segment-list[0] fcbb:bb01:300::
39         Gateway opaque handle: 0x5f6e06e04c20
40         Next hop type: Router, Next hop index: 0
41         Address: 0x5f6e06f72b9c
42         Next-hop reference count: 14, Next-hop session id: 0
43         Kernel Table Id: 0
44         Next hop: fe80::cccc:22ff:feb4:62f1 via enp8s0
45         Protocol next hop: fcbb:bb01:300::
46         Composite next hop: 0x5f6e09e75800 - INH Session ID: 0, CNH non-key opaque: (nil),
         CNH key opaque: 0x5f6e075b8380
47         Indirect next hop: 0x5f6e07159d08 - INH Session ID: 0, INH non-key opaque: 0
         x5f6e06e06e40, INH key opaque: (nil)
48         State: <Secondary Active Int Ext OpaqueData ProtectionCand>
49         Peer AS: 64512
50         Age: 5:07 Metric2: 20
51         Validation State: unverified
52         ORR Generation-ID: 0
53         Task: BGP_64512_64512.3.3.3
54         Announcement bits (4): 2-KRT MFS 3-KRT 4-KRT-vRouter 6-PLFM-LAYER
55         AS path: I
56         Communities: target:64512:4
57         Import Accepted MultiNextHop RecvNextHopIgnored
58         SRv6 SID: fcbb:bb01:300:: Service tlv type: 5 Behavior: 63 BL: 32 NL: 16 FL: 16 AL:
         0 TL: 16 TO: 48
59         VPN Label: 917552
60         Localpref: 100
61         Router ID: 3.3.3.3
62         Primary Routing Table: bgp.l3vpn.0
63         Thread: junos-main
64
65     bgp.l3vpn.0: 1 destinations, 1 routes (1 active, 0 holddown, 0 hidden)
66
67     10.87.3.233:2:30.30.14.11/32 (1 entry, 0 announced)
68         *BGP Preference: 170/-101
69         Route Distinguisher: 10.87.3.233:2
70         Next hop type: Indirect, Next hop index: 0
71         Address: 0x5f6e06f7265c
72         Next-hop reference count: 4
73         Kernel Table Id: 0
74         Source: 3.3.3.3
75         Next hop type: List, Next hop index: 0
76     Next hop: ELNH Address 0x5f6e06f71a1c, selected
77         Next hop type: Chain, Next hop index: 0
78         Address: 0x5f6e06f71a1c
79         Next-hop reference count: 1, Next-hop session id: 0
80         Kernel Table Id: 0
81         Next hop: via Chain Tunnel Composite, SRv6 (src 4444::1 dest fcbb:bb01:300::)
82         Next hop: ELNH Address 0x5f6e06f715bc
83         SRV6-Tunnel: Reduced-SRH Encap-mode Remove-Last-Sid Propagate-CoS
84         Src: 4444::1 Dest: fcbb:bb01:300::
85         Segment-list[0] fcbb:bb01:300::
86         Gateway opaque handle: 0x5f6e06e04c20
87         Next hop type: Router, Next hop index: 0
88         Address: 0x5f6e06f715bc
89         Next-hop reference count: 14, Next-hop session id: 0

```

```

90      Kernel Table Id: 0
91      Next hop: fe80::5054:ff:feab:aecc via enp10s0
92      Next hop: ELNH Address 0x5f6e06f7655c
93      Next hop type: Chain, Next hop index: 0
94      Address: 0x5f6e06f7655c
95      Next-hop reference count: 1, Next-hop session id: 0
96      Kernel Table Id: 0
97      Next hop: via Chain Tunnel Composite, SRv6 (src 4444::1 dest fcbb:bb01:300::)
98      Next hop: ELNH Address 0x5f6e06f72b9c
99      SRV6-Tunnel: Reduced-SRH Encap-mode Remove-Last-Sid Propagate-CoS
100     Src: 4444::1 Dest: fcbb:bb01:300::
101     Segment-list[0] fcbb:bb01:300::
102     Gateway opaque handle: 0x5f6e06e04c20
103     Next hop type: Router, Next hop index: 0
104     Address: 0x5f6e06f72b9c
105     Next-hop reference count: 14, Next-hop session id: 0
106     Kernel Table Id: 0
107     Next hop: fe80::cccc:22ff:feb4:62f1 via enp8s0
108     Protocol next hop: fcbb:bb01:300::
109     Composite next hop: 0x5f6e09e75800 - INH Session ID: 0, CNH non-key opaque: (nil),
110     CNH key opaque: 0x5f6e075b8380
111     Indirect next hop: 0x5f6e07159d08 - INH Session ID: 0, INH non-key opaque: 0
112     x5f6e06e06e40, INH key opaque: (nil)
113     State: <Active Int Ext ProtectionPath ProtectionCand>
114     Peer AS: 64512
115     Age: 5:07 Metric2: 20
116     Validation State: unverified
117     ORR Generation-ID: 0
118     Task: BGP_64512_64512.3.3.3
119     AS path: I
120     Communities: target:64512:4
121     Import Accepted MultiNexthop RecvNextHopIgnored
122     SRv6 SID: fcbb:bb01:300:: Service tlv type: 5 Behavior: 63 BL: 32 NL: 16 FL: 16 AL:
123     0 TL: 16 TO: 48
124     VPN Label: 917552
125     Localpref: 100
126     Router ID: 3.3.3.3
127     Secondary Tables: srv6.inet.0
128     Thread: junos-main

```

### 1.5.11 Remote VPN route on PE1 : Data Path

```

1  bash-5.1# rt --get 30.30.24.11/32 --vrf 2
2  Match 30.30.24.11/32 in vRouter inet4 table 0/2/unicast
3
4  Flags: L=Label Valid, P=Proxy ARP, T=Trap ARP, F=Flood ARP, Ml=MAC-IP learnt route
5  vRouter inet4 routing table 0/2/unicast
6  Destination PPL Flags Label Nexthop Stitched MAC(Index)
7  30.30.24.11/32 0 PT - 59 -
8  bash-5.1# nhchain --get 59
9  Id:59 Type:Indirect Fmly: AF_INET Rid:0 Ref_cnt:2 Vrf:0
10     Next NH:57 NH Label:0 NH Hit Count:1396867
11     Flags:Valid, Etree Root,
12
13  Id:57 Type:Composite Fmly: AF_INET Rid:0 Ref_cnt:2 Vrf:0
14     Next NH:-1 NH Label:0 NH Hit Count:1396867
15     Flags:Valid, Policy, Ecmp, Etree Root,
16     Valid Hash Key Parameters: Proto,SrcIP,SrcPort,DstIp,DstPort
17     Sub NH(label): 55 55 55
18
19  bash-5.1# nhchain --get 57
20  Id:57 Type:Composite Fmly: AF_INET Rid:0 Ref_cnt:2 Vrf:0
21     Next NH:-1 NH Label:0 NH Hit Count:1649059
22     Flags:Valid, Policy, Ecmp, Etree Root,
23     Valid Hash Key Parameters: Proto,SrcIP,SrcPort,DstIp,DstPort
24     Sub NH(label): 55 55 55
25
26  Id:55 Type:Tunnel Fmly:AF_INET6 Rid:0 Ref_cnt:4 Vrf:0
27     Next NH:53 NH Label:0 NH Hit Count:1649059
28     Flags:Valid, Policy, Etree Root, Underlay Ecmp, SRv6,
29     Oif:2 EncapValid:0 Len:0 Data:NULL
30     Oif:0 EncapValid:0 Len:0 Data:NULL
31     Oif:0 EncapValid:0 Len:0 Data:NULL
32     Oif:0 EncapValid:0 Len:0 Data:NULL
33     Oif:0 EncapValid:0 Len:0 Data:NULL
34     Oif:0 EncapValid:0 Len:0 Data:NULL
35     Oif:0 EncapValid:0 Len:0 Data:NULL

```

```

36      Oif:0 EncapValid:0 Len:0 Data:NULL
37
38      Sip: 3333::1
39      Block Len:32 Block: fcbb:bb01::
40      Number of Containers:1
41      Container Dips:[1]: fcbb:bb01:400:e003::
42
43 Id:53 Type:Composite Fmly: AF_INET Rid:0 Ref_cnt:5 Vrf:0
44   Next NH:-1 NH Label:0 NH Hit Count:1649059
45   Flags:Valid, Policy, Ecmp, Etree Root,
46   Valid Hash Key Parameters: Proto,SrcIP,SrcPort,DstIp,DstPort
47   Sub NH(label): 39 43 47
48
49 Id:55 Type:Tunnel Fmly:AF_INET6 Rid:0 Ref_cnt:4 Vrf:0
50   Next NH:53 NH Label:0 NH Hit Count:1649059
51   Flags:Valid, Policy, Etree Root, Underlay Ecmp, SRv6,
52   Oif:2 EncapValid:0 Len:0 Data:NULL
53   Oif:0 EncapValid:0 Len:0 Data:NULL
54   Oif:0 EncapValid:0 Len:0 Data:NULL
55   Oif:0 EncapValid:0 Len:0 Data:NULL
56   Oif:0 EncapValid:0 Len:0 Data:NULL
57   Oif:0 EncapValid:0 Len:0 Data:NULL
58   Oif:0 EncapValid:0 Len:0 Data:NULL
59   Oif:0 EncapValid:0 Len:0 Data:NULL
60
61   Sip: 3333::1
62   Block Len:32 Block: fcbb:bb01::
63   Number of Containers:1
64   Container Dips:[1]: fcbb:bb01:400:e003::
65
66 Id:53 Type:Composite Fmly: AF_INET Rid:0 Ref_cnt:5 Vrf:0
67   Next NH:-1 NH Label:0 NH Hit Count:1649059
68   Flags:Valid, Policy, Ecmp, Etree Root,
69   Valid Hash Key Parameters: Proto,SrcIP,SrcPort,DstIp,DstPort
70   Sub NH(label): 39 43 47
71
72 Id:55 Type:Tunnel Fmly:AF_INET6 Rid:0 Ref_cnt:4 Vrf:0
73   Next NH:53 NH Label:0 NH Hit Count:1649059
74   Flags:Valid, Policy, Etree Root, Underlay Ecmp, SRv6,
75   Oif:2 EncapValid:0 Len:0 Data:NULL
76   Oif:0 EncapValid:0 Len:0 Data:NULL
77   Oif:0 EncapValid:0 Len:0 Data:NULL
78   Oif:0 EncapValid:0 Len:0 Data:NULL
79   Oif:0 EncapValid:0 Len:0 Data:NULL
80   Oif:0 EncapValid:0 Len:0 Data:NULL
81   Oif:0 EncapValid:0 Len:0 Data:NULL
82   Oif:0 EncapValid:0 Len:0 Data:NULL
83
84   Sip: 3333::1
85   Block Len:32 Block: fcbb:bb01::
86   Number of Containers:1
87   Container Dips:[1]: fcbb:bb01:400:e003::
88
89 Id:53 Type:Composite Fmly: AF_INET Rid:0 Ref_cnt:5 Vrf:0
90   Next NH:-1 NH Label:0 NH Hit Count:1649059
91   Flags:Valid, Policy, Ecmp, Etree Root,
92   Valid Hash Key Parameters: Proto,SrcIP,SrcPort,DstIp,DstPort
93   Sub NH(label): 39 43 47

```

### 1.5.12 Remote VPN route on PE2 : Data Path

```

1  bash-5.1# rt --get 30.30.14.11/32 --vrf 1
2  Match 30.30.14.11/32 in vRouter inet4 table 0/2/unicast
3
4  Flags: L=Label Valid, P=Proxy ARP, T=Trap ARP, F=Flood ARP, Ml=MAC-IP learnt route
5  vRouter inet4 routing table 0/2/unicast
6  Destination PPL Flags Label Nexthop Stitched MAC(Index)
7  30.30.14.11/32 0 PT - 46 -
8  bash-5.1# nhchain --get 46
9  Id:46 Type:Indirect Fmly: AF_INET Rid:0 Ref_cnt:2 Vrf:0
10   Next NH:44 NH Label:0 NH Hit Count:0
11   Flags:Valid, Etree Root,
12
13 Id:44 Type:Composite Fmly: AF_INET Rid:0 Ref_cnt:2 Vrf:0
14   Next NH:-1 NH Label:0 NH Hit Count:0
15   Flags:Valid, Policy, Ecmp, Etree Root,
16   Valid Hash Key Parameters: Proto,SrcIP,SrcPort,DstIp,DstPort

```

```

17      Sub NH(label): 42 42
18
19  bash-5.1# nhchain --get 44
20  Id:44 Type:Composite Fmly: AF_INET Rid:0 Ref_cnt:2 Vrf:0
21      Next NH:-1 NH Label:0 NH Hit Count:0
22      Flags:Valid, Policy, Ecmp, Etree Root,
23      Valid Hash Key Parameters: Proto,SrcIP,SrcPort,DstIp,DstPort
24      Sub NH(label): 42 42
25
26  Id:42 Type:Tunnel Fmly:AF_INET6 Rid:0 Ref_cnt:3 Vrf:0
27      Next NH:38 NH Label:0 NH Hit Count:0
28      Flags:Valid, Policy, Etree Root, Underlay Ecmp, SRv6,
29      Oif:1 EncapValid:0 Len:0 Data:NULL
30      Oif:0 EncapValid:0 Len:0 Data:NULL
31      Oif:0 EncapValid:0 Len:0 Data:NULL
32      Oif:0 EncapValid:0 Len:0 Data:NULL
33      Oif:0 EncapValid:0 Len:0 Data:NULL
34      Oif:0 EncapValid:0 Len:0 Data:NULL
35      Oif:0 EncapValid:0 Len:0 Data:NULL
36      Oif:0 EncapValid:0 Len:0 Data:NULL
37
38      Sip: 4444::1
39      Block Len:32 Block: fcbb:bb01::
40      Number of Containers:1
41      Container Dips:[1]: fcbb:bb01:300:e003::
42
43  Id:38 Type:Composite Fmly: AF_INET Rid:0 Ref_cnt:5 Vrf:0
44      Next NH:-1 NH Label:0 NH Hit Count:0
45      Flags:Valid, Policy, Ecmp, Etree Root,
46      Valid Hash Key Parameters: Proto,SrcIP,SrcPort,DstIp,DstPort
47      Sub NH(label): 36 35
48
49  Id:42 Type:Tunnel Fmly:AF_INET6 Rid:0 Ref_cnt:3 Vrf:0
50      Next NH:38 NH Label:0 NH Hit Count:0
51      Flags:Valid, Policy, Etree Root, Underlay Ecmp, SRv6,
52      Oif:1 EncapValid:0 Len:0 Data:NULL
53      Oif:0 EncapValid:0 Len:0 Data:NULL
54      Oif:0 EncapValid:0 Len:0 Data:NULL
55      Oif:0 EncapValid:0 Len:0 Data:NULL
56      Oif:0 EncapValid:0 Len:0 Data:NULL
57      Oif:0 EncapValid:0 Len:0 Data:NULL
58      Oif:0 EncapValid:0 Len:0 Data:NULL
59      Oif:0 EncapValid:0 Len:0 Data:NULL
60
61      Sip: 4444::1
62      Block Len:32 Block: fcbb:bb01::
63      Number of Containers:1
64      Container Dips:[1]: fcbb:bb01:300:e003::
65
66  Id:38 Type:Composite Fmly: AF_INET Rid:0 Ref_cnt:5 Vrf:0
67      Next NH:-1 NH Label:0 NH Hit Count:0
68      Flags:Valid, Policy, Ecmp, Etree Root,
69      Valid Hash Key Parameters: Proto,SrcIP,SrcPort,DstIp,DstPort
70      Sub NH(label): 36 35

```

### 1.5.13 VRF translation next-hop on PE2

Packet received from PE1 at PE2 does a VRF lookup which is represented by this next-hop. PE2 removes SRv6 header and maps the SID to a VRF and a route lookup is done on inner packet.

```

1  bash-5.1# rt --get fcbb:bb01:400:e003::/128 --vrf 0 --family inet6
2  Match fcbb:bb01:400:e003::/128 in vRouter inet6 table 0/0/unicast
3
4  Flags: L=Label Valid, P=Proxy ARP, T=Trap ARP, F=Flood ARP, Ml=MAC-IP learnt route
5  vRouter inet6 routing table 0/0/unicast
6  Destination PPL Flags Label Nexthop Stitched MAC(Index)
7  fcbb:bb01:400:e003::/64 0 T - 32 -
8
9  bash-5.1# nhchain --get 32
10 Id:32 Type:Vrf_Translate Fmly:AF_INET6 Rid:0 Ref_cnt:7 Vrf:1
11     Next NH:-1 NH Label:0 NH Hit Count:260799041
12     Flags:Valid, Etree Root, SRv6,
13     Vrf:1

```



### 1.5.14 Local route on PE2

```

1 bash-5.1# rt --get 30.30.24.11/32 --vrf 1
2 Match 30.30.24.11/32 in vRouter inet4 table 0/1/unicast
3
4 Flags: L=Label Valid, P=Proxy ARP, T=Trap ARP, F=Flood ARP, Ml=MAC-IP learnt route
5 vRouter inet4 routing table 0/1/unicast
6 Destination PPL Flags Label Nexthop Stitched MAC(Index)
7 30.30.24.11/32 0 PT - 34 -
8 bash-5.1# nhchain --get 34
9 Id:34 Type:Encap Fmly:AF_INET/6 Rid:0 Ref_cnt:3 Vrf:1
10     Next NH:-1 NH Label:0 NH Hit Count:265986309
11     Flags:Valid, Policy, Etree Root,
12     EncapFmly:0806 Oif:8 Len:14
13     Encap Data: aa bb cc dd ee 65 00 00 5e 00 01 00

```

### 1.5.15 Packet sent out from PE1 with SRv6 header

```

1 12:40:35.673370 52:54:00:fe:c1:b8 > da:d3:fa:05:0d:38, ethertype IPv6 (0x86dd), length 100: (
2   hlim 64, next-header IPIP (4) payload length: 46) 3333::1 > fcbb:bb01:400:e003::: (tos 0x0,
3   ttl 3, id 3207, offset 0, flags [none], proto UDP (17), length 46)
4 30.30.14.11.1234 > 30.30.24.11.5678: [udp sum ok] UDP, length 18
5   0x0000: dad3 fa05 0d38 5254 00fe c1b8 86dd 6000
6   0x0010: 0000 002e 0440 3333 0000 0000 0000 0000
7   0x0020: 0000 0000 0001 fcbb bb01 0400 e003 0000
8   0x0030: 0000 0000 0000 4500 002e 0c87 0000 0311
9   0x0040: 48e7 1e1e 0e0b 1e1e 180b 04d2 162e 001a
10  0x0050: be9b 6b6c 6d6e 6f70 7172 7374 7576 7778
11  0x0060: 797a 3031

```

### 1.5.16 Configuration on PE1

```

1 set interfaces lo0 unit 0 family inet address 3.3.3.3/32
2 set interfaces lo0 unit 0 family inet6 address 3333::1/128
3 set interfaces lo0 unit 0 family iso address 49.0002.0192.0168.0003.00
4 set interfaces enp7s0 unit 0 family iso
5 set interfaces enp9s0 unit 0 family iso
6 set interfaces enp10s0 unit 0 family iso
7 set routing-options router-id 3.3.3.3
8 set routing-options route-distinguisher-id 3.3.3.3
9 set protocols isis interface lo0.0
10 set protocols isis interface enp7s0 level 2 hello-interval 15
11 set protocols isis interface enp7s0 level 2 hold-time 600
12 set protocols isis interface enp7s0 hello-padding disable
13 set protocols isis interface enp7s0 point-to-point
14 set protocols isis interface enp9s0 level 2 hello-interval 15
15 set protocols isis interface enp9s0 level 2 hold-time 600
16 set protocols isis interface enp9s0 hello-padding disable
17 set protocols isis interface enp9s0 point-to-point
18 set protocols isis interface enp10s0 level 2 hello-interval 15
19 set protocols isis interface enp10s0 level 2 hold-time 600
20 set protocols isis interface enp10s0 hello-padding disable
21 set protocols isis interface enp10s0 point-to-point
22 set protocols isis level 1 disable
23 set protocols isis source-packet-routing srv6 locator u_loc micro-node-sid
24 set routing-options source-packet-routing srv6 block usid_blk_with_statics fcbb:bb01::/32
25 set routing-options source-packet-routing srv6 block usid_blk_with_statics local-micro-sid
   maximum-static-sids 2000
26 set routing-options source-packet-routing srv6 locator u_loc fcbb:bb01:300::/48
27 set routing-options source-packet-routing srv6 locator u_loc micro-sid block-name
   usid_blk_with_statics
28 set routing-options source-packet-routing srv6 locator u_loc micro-sid flavor none
29 set routing-options resolution preserve-nexthop-hierarchy
30 set routing-options transport-class auto-create
31 set routing-options forwarding-table srv6-chain-merge
32 set routing-options forwarding-table export pplb
33 set routing-options forwarding-table channel vrouter export pplb
34 set policy-options policy-statement pplb then load-balance per-packet
35 set system processes routing bgp tcp-listen-port 178
36 set protocols bgp tcp-connect-port 178
37 set protocols bgp group PE_3_4 type internal
38 set protocols bgp group PE_3_4 multihop

```

```

39 set protocols bgp group PE_3_4 local-address 3.3.3.3
40 set protocols bgp group PE_3_4 family inet unicast extended-nexthop
41 set protocols bgp group PE_3_4 family inet unicast advertise-srv6-service
42 set protocols bgp group PE_3_4 family inet unicast accept-srv6-service
43 set protocols bgp group PE_3_4 family inet-vpn unicast extended-nexthop
44 set protocols bgp group PE_3_4 family inet-vpn unicast advertise-srv6-service
45 set protocols bgp group PE_3_4 family inet-vpn unicast accept-srv6-service
46 set protocols bgp group PE_3_4 family inet6 unicast advertise-srv6-service
47 set protocols bgp group PE_3_4 family inet6 unicast accept-srv6-service
48 set protocols bgp group PE_3_4 family inet6-vpn unicast advertise-srv6-service
49 set protocols bgp group PE_3_4 family inet6-vpn unicast accept-srv6-service
50 set protocols bgp group PE_3_4 family evpn signaling
51 set protocols bgp group PE_3_4 local-as 64512
52 set protocols bgp group PE_3_4 neighbor 4.4.4.4
53 set protocols bgp source-packet-routing srv6 locator u_loc micro-dt4-sid
54 set protocols bgp source-packet-routing srv6 locator u_loc micro-dt6-sid
55 set protocols bgp source-packet-routing srv6 locator u_loc micro-dt46-sid
56 set protocols source-packet-routing srv6
57 set routing-instances srv6 protocols bgp source-packet-routing srv6 locator u_loc micro-dt4-sid
58 set routing-instances srv6 protocols bgp source-packet-routing srv6 locator u_loc micro-dt6-sid
59 set routing-instances srv6 protocols bgp source-packet-routing srv6 locator u_loc micro-dt46-sid
60 set groups cni routing-instances srv6 instance-type vrf
61 set groups cni routing-instances srv6 routing-options rib srv6.inet6.0 static route 1234::1e1e:
e0b/128 qualified-next-hop 1234::1e1e:e0b interface vhostnet5-6b7ae4ee-dedd-4410-b3
62 set groups cni routing-instances srv6 routing-options static route 30.30.14.11/32 qualified-next
-hop 30.30.14.11 interface vhostnet5-6b7ae4ee-dedd-4410-b3
63 set groups cni routing-instances srv6 interface vhostnet5-6b7ae4ee-dedd-4410-b3
64 set groups cni routing-instances srv6 vrf-target target:64512:4

```

## 1.6 Acknowledgement

- Author likes to thank Mahesh Sivakumar for his contribution to the design. Together, we designed scalable and extendable SRv6 design for JCNR.
- Author likes to thank Mahesh Sivakumar, Bharath R, Vinay Agrawal and Ved Patel for their partnership in SRv6 with JCNR initiative.

## 1.7 Glossary

- BGP: Border Gateway Protocol
- CE: Customer Edge
- IS-IS: Intermediate System to Intermediate System
- P: Provider
- PE: Provider Edge
- L3VPN: Layer 3 Virtual Private Network
- OSPF: Open Shortest Path First
- PFE: Packet Forwarding Engine
- SID: Segment Identifier
- SRH: Segment Routing Header
- SRv6: Segment Routing version 6
- VRF: Virtual Routing and Forwarding instance

### Useful links

- RFC 8986: Segment Routing over IPv6 (SRv6) Network Programming: <https://datatracker.ietf.org/doc/rfc8986/>
- SRv6 in Junos: <https://www.juniper.net/documentation/us/en/software/junos/is-is/topics/topic-map/infocus-isis-srv6-network-programming.html>



- Segment Routing in JCNR : <https://www.juniper.net/documentation/us/en/software/cloud-native-router25.2/cloud-native-router-user/topics/concept/segment-routing.html>
- JCNR overview : <https://www.juniper.net/documentation/us/en/software/cloud-native-router25.2/cloud-native-router-user/topics/concept/jcncr-overview.html>



DC – AI – Routing – Switching – Security – Wireless – Automation

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